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MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
FOURTH SEMESTER B. TECH. DEGREE EXAMINATION, MONTH AND YEAR

Course Code: PEEET412
Course Name: RENEWABLE ENERGY SOURCES

Max. Marks: 60 Duration: 2 hours 30 minutes

PART A

Answer all questions. Each question carries 3 marks

			CO	Marks
1		How does switching over to renewable energy sources aid the world towards sustainable growth?	1	(3)
2		With a neat diagram give the function of the essential part of a Darrius type wind energy conversion system.	3	(3)
3		How pyrheliometer measures solar radiation?	3	(3)
4		A house owner wants to switch over to stand alone system from grid connected PV system. List out the advantages and disadvantages of both system.	3	(3)
5		Explain the concept of Waste to Energy.	4	(3)
6		A double cycle tidal energy system can harvest more energy than single ebb and single tide system. Justify the statement.	4	(3)
7		Bio fouling is one the main disadvantage of an OTEC system.	4	(3)
8		What are the methods by which the concept of a net-zero energy building can be realized?	2	(3)

PART B

Answer any one full question from each module. Each question carries 9 marks

Module 1

9	a.	Betz's limit is one of the major limitation of a wind energy conversion system. Explain the concept of Betz's limit and derive capacity factor of a WECS with a neat diagram showing expansion of a column of air passing through a turbine.	3	(6)
	b.	What are the major initiative by the Government of India to promote renewable energy resources to rural areas.	2	(3)
		OR		
10		With a neat diagram and labeling all the major parts explain the working of a wind energy conversion system. Clearly specify the functions of each parts of the system,	3	(9)

Module 2

11	a.	Explain the working of a CTC based solar power conversion system	3	(7)
	b.	What is MPPT in connection with PV plants?	3	(2)
		OR		
12	a.	Explain the principles behind working of a solar cell.	3	(6)

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	b.	How solar irradiation is measured in an inclined surface?	3	(3)

Module 3

13	a.	What is the process of gasification to produce syngas? With a neat diagram of a upgraft gasifier explain the zones in a gasifier.	3	(6)
	b.	Define i) Single ebb cycle TECS ii) Single tide cycle TECS	3	(3)

OR

14		Explain the working of a double basin tidal energy conversion system.	3	(9)

Module 4

15	a.	Explain the working of a hybrid OTEC system.	3	(7)
	b.	Why Hydrogen is considered as the fuel for future?	2	(2)

OR

16	a.	Explain the working of a fuel cell with an example of a Phosphoric acid fuel cell.	3	(7)
	b.	Draw the diagram of an Open OTEC system.	3	(2)

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